

Simplify Invoicing by Creating a Template with Python



Businesses require customised templates to reduce the time and effort spent on each document. This article talks about reclaiming the hours spent in creating invoices by designing a template in Python.

All businesses raise invoices to generate funds. I run a consultancy service that raises bills every week, so that my clients don't need to get approval for larger monthly or annual invoices from their finance department. However, a major constraint here is that a lot of time is spent in designing the invoices. There are lots of services like Zoho books that allow you to make templates, but for a few specific cases like mine, the

constraint is the time spent in typing out the actual items for the invoices.

The workflow that is generally used for creating such invoices is as follows.

1. Create a copy of a Google Drive *doc* file that serves as a template.
2. Read GitLab issues for the project, and create a list of items to be added in the invoice.
3. Download the *doc* as a PDF file and email it to the client.

Though this workflow is fine for

1-4 projects at a time, when 8-10 projects are handled, this process starts consuming an hour every week. A lot of time can be saved by using the following steps.

The template

First, a template in *docx* format is needed. It is preferable to have a Google Drive doc as a template and use words as placeholders for actual information; for example : 'BILL_TO', 'INVOICE_DATE', 'ITEM_LIST', and 'TOTAL_AMOUNT'.

The good thing about the *docx* format is that it's actually a zip file. Since Python has a zipfile module (<https://docs.python.org/3/library/zipfile.html>), it's easy to work with *docx* in it for this particular case.

This step simply requires that you download this *docx* template file onto your computer.

Invoices

GitLab can be used to manage projects. The CI/CD pipelines have to be set up to ensure that all merge request titles follow a certain pattern. I used '#<issue number>: Brief description of task' as the template. This template, along with squash commits, makes sure that the main branch only has commits that link to issues and a brief description of what is achieved. All the intermediate commits of 'why is this not working?', 'tests are passing', 'making it faster' get collapsed within the merge request, and the main branch remains clean thanks to squash commits.



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Invoice no	{{ INVOICE_NUMBER }}		
Bill to	{{ BILL_TO }}		
Pay before	{{ PAY_BEFORE }}	Issued on	{{ GEN_ON }}
Payment method	{{ PAY_TO_DETAILS }}		

Sr	Description	Rate	Qty	Amount
{{ SR_NO }}	{{ ITEM_DESCRIPTION }}	{{ ITEM_RATE }}	{{ ITEM_QTY }}	{{ ITEM_AMOUNT }}
Total				{{ TOTAL }}
				{{ TOTAL_WORDS }}

For any clarifications please write an email to us at 

Figure 1: The template